SYRUP PRODUCTION GUIDE

Craft Fruit Syrup Production, Shelf Life, and Stability Techniques

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1. Introduction and Purpose of the Document

Syrup production constitutes an important production category in the beverage industry, gastronomy, and mixology. Fruit-based syrups have a wide range of applications, from cocktail preparation to coffee flavoring, from soda-based beverages to dessert production. In recent years, with the growth of craft beverage culture, interest in natural syrups produced with real fruit has increased significantly.

Craft fruit syrup production is not merely about preparing a sugary liquid. This production approach is based on scientific principles such as preserving the aromatic character of the fruit, ensuring microbial stability, and controlling shelf life.

The purpose of this document is to explain the fundamental technical principles used in craft fruit syrup production. The document focuses particularly on the following topics:

- Basic concepts in syrup production
- Use of fruit juice and concentrate
- Sugar and acid balance
- The concept of Brix
- pH control
- Water activity (Aw)
- Factors determining shelf life
- Pasteurization and hot-fill techniques

This guide has been prepared as a technical handbook for bars, cafés, craft beverage producers, and small-scale syrup manufacturers.

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2. Basic Concepts Related to Syrup Production

A syrup is basically a viscous liquid formed by dissolving sugar in water. In fruit syrups, fruit juice, aroma compounds, and acidity regulators are added to this base mixture.

There are three main goals in syrup production:

1. Aromatic quality
2. Microbial stability
3. Shelf life

Aromatic quality is the ability of the syrup to reflect the true fruit character. This feature is especially important in craft production systems.

Microbial stability refers to the product's resistance to microbial growth. Sugar content, pH value, and water activity play critical roles in achieving this stability.

Shelf life refers to the period during which the product can be stored without spoilage. Shelf life depends not only on the formulation but also on production hygiene, filtration methods, and filling technique.

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3. Fruit Juice or Concentrate?

The fruit raw material used in syrup production can be obtained in two different ways:

1. Fresh fruit juice
2. Fruit concentrate

Both methods have different advantages and disadvantages.

Use of Fresh Fruit Juice

The use of fresh fruit juice is especially preferred in craft production. The main reason for this is the preservation of the real fruit aroma.

Advantages:

- More natural aroma
- More complex flavor profile
- Premium product perception
- Possibility of additive-free production

Disadvantages:

- Microbial load may be higher
- Brix and pH values may vary
- Shelf life control may be more difficult

Use of Fruit Concentrate

Concentrated fruit products are often preferred in industrial production.

Advantages:

- Standard quality
- Stable Brix value
- Long shelf life

Disadvantages:

- Aromatic complexity may be lower
- In some cases, an artificial taste perception may occur

In craft production systems, formulations based on fresh fruit juice are generally preferred.

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4. The Logic of Flavor Balance

The most important factor determining the quality of a syrup is flavor balance. Flavor balance consists of three main components:

- Sugar
- Acid
- Aroma

Combining these three elements in the right proportions makes it possible to obtain a balanced syrup.

Sugar

Sugar does not only provide sweetness. It also performs the following functions:

- Increases viscosity
- Reduces water activity
- Extends shelf life

Acid

Acid prevents sweetness from becoming too heavy and makes the fruit aroma more vibrant.

Commonly used acid sources:

- Lemon juice
- Citric acid
- Malic acid

Aroma

Aroma consists of volatile compounds that define the character of the fruit. Since these compounds can easily be lost at high temperatures, excessive boiling should be avoided during production.

Flavor balance can be simply expressed as follows:

Flavor Balance = Sugar + Acid + Aroma

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5. What Determines the Sugar Ratio?

The amount of sugar to be used in syrup formulation is not determined randomly. Three main parameters should be considered:

1. The fruit's pH value
2. The fruit's Brix value

3. The fruit's aromatic intensity

The Fruit's pH Value

Fruits with lower pH values are more acidic. In such fruits, using less sugar may often be sufficient.

The Fruit's Brix Value

The Brix value indicates the natural sugar content of the fruit. In fruits with high Brix values, the amount of added sugar can be reduced.

Aromatic Intensity

Some fruits have very strong aromas. For example, fruits such as passion fruit or pomegranate can provide intense aroma even at lower sugar levels.

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6. What is Brix?

Brix is a unit of measurement that expresses the amount of dissolved sugar in a liquid.

1 Brix approximately means 1% sugar.

Example:

Brix	Sugar Ratio
10 Brix	10%
50 Brix	50%
60 Brix	60%

The target range in syrup production is generally:

60–65 Brix

This range provides the following advantages:

- Limits microbial growth
- Creates a denser texture
- Extends shelf life

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7. What is pH?

pH is a measure that indicates whether a liquid is acidic or basic.

pH scale:
0–14

pH Range	Description
0–3	Highly acidic
3–5	Acidic
7	Neutral
7+	Basic

The pH value of fruit-based products is generally:

2.5–4.0

Ideal range for syrup stability:

pH 3.2–3.5

Within this range, many harmful bacteria cannot grow.

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8. What is Aw?

Aw stands for **Water Activity**.

Water activity refers to the amount of free water in a product that microorganisms can use.

The Aw value ranges between 0 and 1.

Aw Value	Microbial Risk
0.98	Very high
0.90	High
0.80	Moderate
0.70	Low

Adding sugar binds part of the water and lowers the Aw value.

For this reason, syrups with high sugar content are more resistant to microorganisms.

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9. Factors Determining Shelf Life

The shelf life of a syrup does not depend solely on the amount of sugar used. There are multiple technical factors that determine shelf life. Controlling these factors together is critical for product stability.

The main factors determining syrup shelf life are as follows:

- Sugar ratio
- pH value

- Water activity (Aw)
- Hygiene conditions
- Filtration
- Pasteurization
- Filling method

Sugar Ratio

A high sugar ratio makes it difficult for microorganisms to grow. This is because sugar lowers water activity. Microorganisms need free water to multiply. Sugar binds some of the water and reduces the amount of water available to microorganisms.

pH Value

A low pH value inhibits the growth of many bacteria. In fruit-based syrups, a pH range of **3.0–3.5** is generally targeted.

Water Activity (Aw)

As the Aw value decreases, the likelihood of microbial growth also decreases.

Hygiene

The cleanliness of the equipment used during production has a direct effect on shelf life.

Filtration

Filtration reduces pulp and microbial load.

Pasteurization

Heat treatment inactivates a large portion of microorganisms.

Filling Method

The hot-fill method extends shelf life by reducing microbial load inside the bottle.

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10. The Effect of High Sugar Content

The sugar ratio used in syrup production directly affects product stability.

There are two main systems in syrup production:

- 1:1 Simple Syrup
- 2:1 Rich Syrup

Simple Syrup (1:1)

Formula:

1 part sugar
1 part water

This system is generally used in syrups for beverages that are consumed quickly.

Average characteristics:

- Brix: around 50
- Aw: higher
- Shelf life: shorter

Rich Syrup (2:1)

Formula:

2 parts sugar
1 part water

This system creates a more stable structure.

Average characteristics:

- Brix: 60–65
- Aw: lower
- Shelf life: longer

Comparison Table

Syrup Type	Brix	Shelf Life (refrigerated)
Simple Syrup	45–50	2–3 weeks
Fruit Simple Syrup	45–50	1–2 weeks
Rich Syrup	60–65	1–1.5 months
Fruit Rich Syrup	60–65	3–5 weeks

Syrups containing fruit juice generally have a shorter shelf life.

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11. Use of Natural Acids

The use of acid in syrup production is important both for flavor balance and microbial stability.

The most commonly used natural acid sources are:

- Lemon juice
- Citric acid
- Malic acid

Lemon Juice

Lemon juice is a natural source of acidity and lowers the pH value. It also contributes to aroma.

Citric Acid

Citric acid is especially preferred in industrial production. It imitates natural fruit acids.

Effects of Acid Use

- Lowers pH
- Makes microbial growth more difficult
- Improves flavor balance

When the pH is kept within the **3.2–3.5** range, shelf life can be extended significantly.

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12. Pasteurization

Pasteurization is a controlled heat treatment applied to inactivate microorganisms.

In syrup production, the following temperature range is generally used:

75–85 °C

The purposes of pasteurization are:

- To reduce yeasts and molds
- To lower bacterial load
- To extend shelf life

Pasteurization Procedure

1. The syrup mixture is prepared
2. The mixture is heated to around **80 °C**
3. A homogeneous mixture is ensured

This process is especially important in fruit-containing syrups.

Shelf Life After Pasteurization

Product Type	Shelf Life
Fruit Simple Syrup	2–3 weeks
Fruit Rich Syrup	3–5 weeks
Pasteurized Rich Syrup	1–2 months

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13. Hot Fill

Hot filling is the process of bottling the product at a high temperature after pasteurization.

This method provides the following advantages:

- Elimination of microorganisms inside the bottle

- Sterility of the cap area
- Extended shelf life

Hot Fill Procedure

1. The syrup is kept at around **80 °C**
2. It is filled into sterile bottles
3. The cap is closed immediately
4. The bottle is briefly inverted

This method provides very effective stability, especially in small-scale production.

Shelf Life After Hot Fill

Syrup Type	Shelf Life
Simple Syrup	3–4 weeks
Fruit Rich Syrup	4–6 weeks
Acid-balanced Rich Syrup	1.5–2 months

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14. Filtration and Hygiene

Filtration and hygiene are among the most critical factors in syrup production.

During production, fruit pulp and particles may remain in the syrup. These particles can create a suitable environment for microorganisms.

For this reason, filtration is recommended after production.

Filtration Methods

- Fine strainer
- Cloth filter
- Coffee filter
- Paper filter

Benefits of Filtration

- Clearer product
- Lower microbial load
- Longer shelf life

From a hygiene perspective, the following rules should be applied:

- Equipment must be disinfected before production
- Bottles must be sterile

- The working area must be kept clean

When hygiene conditions are poor, shelf life can decrease significantly.

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15. Average pH Values of Fruits

The following table shows the average pH values of some common fruits.

Fruit	Average pH
Lemon	2.0
Lime	2.1
Pomegranate	3.0
Passion fruit	3.0
Strawberry	3.3
Sour cherry	3.2
Orange	3.5
Pineapple	3.4
Mango	3.8
Peach	3.6
Banana	4.5
Melon	6.0
Watermelon	5.5

Fruits with lower pH values are generally more suitable for producing more stable syrups.

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16. Average Brix Values of Fruits

The following table shows the average Brix values of some fruits.

Fruit	Average Brix
Lemon	6
Lime	7
Pomegranate	14
Passion fruit	13

Fruit	Average Brix
Strawberry	8
Sour cherry	12
Orange	11
Pineapple	14
Mango	15
Peach	12
Banana	18
Melon	10
Watermelon	9

Fruits with higher Brix values have higher natural sugar content.

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17. Sugar Addition by Fruit Type

The amount of sugar to be used in syrup production is not determined solely by the target Brix value. The fruit's natural acidity, aromatic intensity, and natural sugar content also play an important role in the formulation.

In terms of syrup production, fruits can generally be divided into five categories.

Highly Sour Fruits

These fruits have high acidity and generally low Brix values.

Example fruits:

- Lemon
- Lime
- Passion fruit

Since acidity is quite high in these fruits, the sugar amount can be kept higher.

Recommended formulation:

- 1 liter of fruit juice
- 700–900 g sugar

These syrups have strong aroma and provide intense character in cocktails.

Sour Fruits

These fruits have noticeable acidity but do not contain an extremely sharp sourness.

Example fruits:

- Sour cherry
- Pomegranate
- Pineapple

Recommended formulation:

- 1 liter of fruit juice
- 600–800 g sugar

This category provides a very balanced structure for cocktail syrups.

Balanced Fruits

These fruits have a more stable sugar-acid balance.

Example fruits:

- Orange
- Strawberry
- Peach

Recommended formulation:

- 1 liter of fruit juice
- 500–700 g sugar

This category is generally the easiest fruit group to formulate.

Sweet Fruits

Fruits with high natural sugar content fall into this category.

Example fruits:

- Mango
- Melon

Recommended formulation:

- 1 liter of fruit juice
- 400–600 g sugar

Excessive sugar use in these fruits may disrupt flavor balance.

Very Sweet Fruits

These fruits have high Brix values and low acidity.

Example fruits:

- Banana
- Watermelon

Recommended formulation:

- 1 liter of fruit juice
- 300–500 g sugar

In these fruits, additional acid is generally required.

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18. Sugar Guide by pH Range

In syrup production, the amount of sugar can be adjusted according to the fruit's pH value. Fruits with lower pH values are more acidic and can therefore tolerate higher sugar ratios.

The table below shows the average sugar amount according to pH value.

pH Range	Fruit Acidity	Recommended Sugar Amount
2.0–2.5	Very high acid	800–1000 g / liter
2.5–3.0	High acid	700–900 g / liter
3.0–3.5	Medium acid	600–800 g / liter
3.5–4.0	Low acid	500–700 g / liter
4.0–5.0	Very low acid	400–600 g / liter

This table can be used as a practical reference for formulation planning.

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19. Aw Reduction Techniques

Aw, or water activity, refers to the amount of free water required for microorganisms to grow. Reducing the Aw value can significantly extend the shelf life of syrup.

The main methods used to lower Aw are as follows.

Use of Sugar

Sugar binds part of the water and reduces the amount of water that microorganisms can use. For this reason, syrups with high sugar content are more stable.

Invert Sugar

Invert sugar is obtained by breaking sucrose down into glucose and fructose.

Advantages:

- More stable structure
- Reduces crystallization
- Lowers Aw value

Glucose Syrup

Glucose syrup is especially used in industrial production. It increases viscosity and lowers water activity.

Alcohol

In some special syrups, low levels of alcohol can be used to increase microbial stability. However, this method is generally preferred in liqueur production.

Lowering pH

A low pH value makes microbial growth more difficult. Therefore, when acidity control is used together with Aw control, it provides more effective results.

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20. Shelf Life Scenarios

The shelf life of syrup varies depending on the formulation and production method used. The following tables show average shelf life estimates for different production scenarios.

Without Pasteurization

Product Type	Shelf Life
Simple syrup	2–3 weeks
Fruit simple syrup	1–2 weeks
Fruit rich syrup	2–3 weeks

With Pasteurization

Product Type	Shelf Life
Simple syrup	3–4 weeks
Fruit rich syrup	3–5 weeks

Product Type	Shelf Life
Acid-balanced rich syrup	4–6 weeks

Pasteurization + Hot Fill

Product Type	Shelf Life
Rich syrup	1–1.5 months
Fruit rich syrup	1–1.5 months
Acid-balanced fruit syrup	1.5–2 months

These values apply to refrigerated storage conditions.

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21. Microorganisms

The microorganisms that generally cause spoilage in syrup production are yeasts and molds.

The main microorganisms that can develop in syrup are as follows.

Yeast

Yeasts can multiply rapidly in sugary environments.

Symptoms:

- Gas formation
- Foaming
- Bottle pressure

Mold

Molds usually develop on the surface.

Symptoms:

- White or green layer on the surface
- Bad odor

Acid Bacteria

Some bacteria may spoil the taste of the syrup.

Symptoms:

- Sour odor

- Cloudiness

Methods to Reduce Microbial Risk

- High sugar ratio
- Low pH
- Pasteurization
- Hot fill
- Hygiene

Applying these factors together significantly extends shelf life.

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22. Basic Syrup Recipes

Some basic recipes form the foundation of many different syrup products. These recipes are considered standard formulas in mixology and beverage production.

The following recipes have been prepared for small-scale craft production.

Simple Syrup

Simple syrup is the most basic type of syrup in cocktail and beverage production. It is prepared by dissolving equal proportions of sugar and water.

Formula

- 500 ml water
- 500 g sugar

Production

1. Water is heated in a saucepan.
2. Sugar is added.
3. The mixture is stirred until completely dissolved.
4. Boiling is not necessary.

Properties

Parameter	Value
Brix	45–50
Shelf life	2–3 weeks

Rich Syrup

Rich syrup is a denser syrup type and has a longer shelf life than simple syrup.

Formula

- 330 ml water
- 660 g sugar

Production

1. Water is heated.
2. Sugar is added.
3. The mixture is stirred until completely dissolved.

Properties

Parameter	Value
Brix	60–65
Shelf life	4–6 weeks

Strawberry Syrup

Strawberry syrup is commonly used in cocktails and soda-based drinks.

Formula

- 500 ml strawberry juice
- 450 g sugar
- 20 ml lemon juice

Production

1. Strawberry juice is prepared.
2. Sugar is added and mixed.
3. The mixture is heated to around **70–80 °C**.
4. Lemon juice is added.

Properties

Parameter	Value
Brix	60
Shelf life	3–4 weeks

Grenadine

Grenadine is a classic pomegranate-based cocktail syrup.

Formula

- 500 ml pomegranate juice
- 500 g sugar
- 15 ml lemon juice

Production

1. Pomegranate juice and sugar are mixed.
2. The mixture is heated to around **75 °C**.
3. Lemon juice is added.

Properties

Parameter	Value
Brix	60
Shelf life	4–5 weeks

Vanilla Syrup

Vanilla syrup is widely used in coffee and cocktails.

Formula

- 500 ml simple syrup
- 1 vanilla bean

Production

1. The vanilla bean is split lengthwise.
2. It is added to the syrup.
3. Infusion is carried out for 24 hours.

Properties

Parameter	Value
Brix	50
Shelf life	3–4 weeks

Blue Curaçao-Type Syrup

Blue Curaçao-type syrups are orange-flavored syrups.

Formula

- 500 ml rich syrup
- 10 ml orange peel extract
- food coloring (optional)

Production

1. Rich syrup is prepared.
2. Orange flavor is added.
3. The mixture is homogenized.

Properties

Parameter	Value
Brix	60
Shelf life	4–6 weeks

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23. Craft Fruit Syrup Production Formula

Using a standard base formula in craft production is beneficial for production stability.

1 Liter of Craft Fruit Syrup

- 500 ml fruit juice
- 450 g sugar
- 30 ml water
- 20 ml lemon juice

Production Steps

1. Fruit juice is prepared.
2. Sugar is added and dissolved.
3. The mixture is heated to **75–80 °C**.
4. Lemon juice is added.
5. Filtration is applied.
6. Hot filling is carried out.

This formula can be used as a basic reference for most fruits.

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24. Production Standard for Formül Atölyesi

Establishing a brand standard in craft syrup production is important in terms of quality. In craft production-focused brands such as Formül Atölyesi, the main approach is the **real fruit syrup** concept.

The real fruit syrup approach is based on the following principles:

- Use of real fruit
- No artificial flavoring
- Balanced sugar ratio
- Controlled pH

Production Standard

1. Real fruit juice should be used.
2. The target Brix value should be around **60**.
3. The pH value should be kept within the **3.2–3.5** range.
4. The product should be pasteurized.
5. Hot filling should be applied.
6. The product should be filtered.

This production standard is sufficient for producing stable and aromatic syrups.

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25. Summary Rules

The basic rules to be considered in craft fruit syrup production are summarized below.

1. Sugar ratio is important for stability.
2. The Brix value should generally be kept around **60**.
3. The pH value should be within the **3.2–3.5** range.
4. Filtration increases product clarity.
5. Pasteurization extends shelf life.
6. Hot filling reduces microbial risk.
7. Hygiene is the most critical element of production.

Applying these principles together makes it possible to produce high-quality and stable craft syrups.

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26. Ideal pH and Sugar Balance by Fruit Type

Different fruits have different acidity and natural sugar levels. Therefore, using the same sugar formula for every fruit is not correct. The table below shows the average pH values of commonly used fruits and the recommended sugar amounts in craft syrup production.

Fruit	Average pH	Natural Brix	Recommended Sugar (for 1 L fruit juice)	Target Brix
Lemon	2.0	6	800–900 g	60–65
Lime	2.1	7	800–900 g	60–65
Pomegranate	3.0	14	700–800 g	60
Passion fruit	3.0	13	700–800 g	60
Sour cherry	3.2	12	650–750 g	60
Strawberry	3.3	8	600–700 g	60
Orange	3.5	11	550–650 g	58–60
Pineapple	3.4	14	550–650 g	58–60
Peach	3.6	12	500–600 g	55–58
Mango	3.8	15	450–550 g	55–58
Melon	6.0	10	450–550 g	55
Watermelon	5.5	9	400–500 g	55

This table can be used as a quick reference during formula development. Fruits with lower pH values can generally tolerate higher sugar levels.

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27. Craft Syrup Formula Calculation Table

The most important calculation method used in syrup production is reaching the target Brix value. For this reason, producers generally calculate sugar quantity according to volume.

The table below shows approximate sugar ratios for producing 1 liter of syrup.

Target Brix	Sugar Amount	Water / Fruit Juice
45 Brix	450 g	550 ml
50 Brix	500 g	500 ml
55 Brix	550 g	450 ml
60 Brix	600 g	400 ml
65 Brix	650 g	350 ml

In craft fruit syrups, a target of around **60 Brix** is generally preferred. This value offers a good balance in terms of both flavor and shelf life.

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28. Syrup Stability Control Checklist

In professional production, it is recommended to check certain parameters after each batch.

Parameters to be checked:

- Brix measurement (with refractometer)
- pH measurement
- Visual clarity
- Aroma check
- Bottle seal integrity

Sample Stability Control Table

Parameter	Target Value
Brix	58–65
pH	3.2–3.5
Aw	< 0.85
Filling temperature	75–80 °C

Checking these parameters significantly improves product stability.

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29. Common Mistakes in Craft Syrup Production

Some common mistakes encountered in syrup production can seriously affect product quality.

Low Sugar Ratio

A sugar ratio that is too low facilitates microbial growth.

High pH

A pH value above **4.0** may accelerate bacterial growth.

Insufficient Filtration

Pulp and particles may create a suitable environment for microbial growth.

Excessive Boiling

Syrups that are boiled for too long may suffer aroma loss.

Poor Hygiene

Unclean production equipment can seriously shorten shelf life.

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30. The Future of Craft Syrup Production

In recent years, with the growth of the craft beverage sector, natural syrup production has also shown significant development. Products made with real fruit are increasingly preferred by both bars and coffee shops.

The following trends stand out in modern craft production:

- Use of real fruit
- Low additive content
- Aromatic diversity
- Small-batch production

This approach both improves product quality and is perceived by consumers as a more natural alternative.

At this point, the guide has reached the level of an approximately 20+ page technical document.

If you wish, I can also write the following for you as a final addition:

“Professional Craft Syrup Production Flow Diagram”

This is something commonly found in books, and if you place it at the beginning of the PDF, it will make the document look much more professional.

Let’s continue.

Great, then we are now writing the final sections that complete the guide like a real technical handbook. In this section, the production process will be presented in flow diagram logic, followed by quality control and production planning sections.

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31. Craft Syrup Production Flow Process

Craft fruit syrup production should be carried out according to a defined production flow. Standardizing this flow is important for maintaining production quality.

The following production process is a basic flow model recommended for small-scale craft producers.

Production Flow Steps

1. Fruit selection
2. Fruit preparation
3. Juice extraction
4. Sugar addition
5. Heat treatment (pasteurization)
6. Acid balance adjustment
7. Filtration

8. Hot filling
9. Cooling
10. Labeling and storage

This flow model helps maintain control at every stage of production.

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32. Fruit Selection and Raw Material Quality

The quality of the fruit used in syrup production directly affects the aromatic character of the product.

The following criteria should be considered for ideal fruit selection:

- Ripeness level
- Aroma intensity
- Absence of rot or mold
- Fresh harvest

Overripe fruits may sometimes provide more intense aroma, but their microbial load may also be higher.

For this reason, fruits should be carefully inspected before production.

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33. Fruit Preparation

Fruits should be properly prepared before production.

The preparation stage generally includes the following processes:

- Washing
- Sorting
- Peeling (if necessary)
- Cutting

Since soil and microorganisms on the fruit surface can cause contamination during production, washing is important.

In industrial production, chlorinated water or ozonated water may be used for washing.

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34. Juice Extraction

Juice production can be carried out using different methods depending on the fruit used.

Common methods

- Pressing

- Blending
- Cold pressing
- Centrifugation

In craft production, the **blender + filtration** method is generally used.

This method helps preserve aromatic compounds.

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35. Heat Treatment and Mixture Preparation

Once the fruit juice is obtained, the syrup mixture is prepared by mixing it with sugar.

The following steps are followed when preparing the mixture:

1. Fruit juice is placed into the mixing vessel
2. Sugar is added
3. The mixture is heated
4. It is stirred until the sugar is completely dissolved

During heating, the usual target temperature is:

75–80 °C

This temperature both ensures sugar dissolution and reduces microbial load.

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36. Filtration

Filtration is applied to improve the clarity of the syrup and remove particles.

Filtration methods

- Fine strainer
- Cloth filter
- Coffee filter
- Paper filter

Filtration can also extend shelf life because it reduces microbial load.

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37. Hot Filling

Hot filling is the process of bottling the syrup at high temperature.

This method provides the following advantages:

- Elimination of microorganisms on the inner surface of the bottle
- Sterility of the cap area

- Extended shelf life

Hot Fill Procedure

1. The syrup is kept at **75–80 °C**
2. It is filled into sterile bottles
3. The cap is closed immediately
4. The bottle is briefly inverted

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38. Cooling and Storage

After filling, the bottles are left to cool at room temperature.

Storage conditions directly affect shelf life.

Recommended storage conditions

- Cool environment
- Away from direct sunlight
- Refrigerator temperature of **4–10 °C**

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39. Labeling

In professional production, product labels should contain certain information.

The following information is recommended on the label:

- Product name
- Ingredient list
- Production date
- Expiry date
- Storage conditions

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40. Quality Control

In every production batch, some basic quality control tests should be carried out.

Recommended parameters to measure

Parameter	Measurement Method
Brix	Refractometer
pH	pH meter

Parameter	Measurement Method
Appearance	Visual inspection
Aroma	Sensory analysis

These controls help maintain the production standard.

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41. Small-Scale Production Planning

In craft production, manufacturing is generally carried out in small batches.

Recommended production quantities

Production Type	Batch Size
Test production	1–2 liters
Small batch	5–10 liters
Small commercial production	20–50 liters

Small-batch production makes quality control easier.

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42. Conclusion

Craft fruit syrup production makes it possible to obtain stable products with high aromatic quality when the correct technical principles are applied.

For successful syrup production, the following factors must be controlled together:

- Correct sugar ratio
- Appropriate pH value
- Water activity control
- Hygiene
- Pasteurization
- Hot filling

Applying these principles makes it possible to produce high-quality craft syrups with an extended shelf life.

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43. Syrup Stability and Maximum Shelf Life Table

Syrup shelf life varies depending on formulation, production method, and storage conditions. The following table shows average shelf life estimates according to different production scenarios.

These values apply to sealed bottles and hygienic production conditions.

Syrup Type	Brix	pH	Production Method	Storage	Shelf Life
Simple syrup	45–50	5–6	No heating	Refrigerator	2–3 weeks
Fruit simple syrup	45–50	3–4	No heating	Refrigerator	1–2 weeks
Rich syrup	60–65	5–6	Heating	Refrigerator	1–1.5 months
Fruit rich syrup	60–65	3–4	Heating	Refrigerator	3–5 weeks
Acid-balanced rich syrup	60–65	3.2–3.5	Pasteurization	Refrigerator	2–3 months
Pasteurized fruit syrup	60–65	3.0–3.5	Pasteurization + hot fill	Refrigerator	3–4 months
High-sugar craft syrup	65–68	3.0–3.3	Pasteurization + hot fill	Refrigerator	4–6 months
Concentrated syrup	70+	3.0–3.5	Pasteurization + hot fill	Refrigerator	6 months

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Frozen Storage

When syrups are stored frozen, shelf life can be extended even further.

Syrup Type	Storage	Shelf Life
Fruit syrup	Freezer	4–6 months
Rich syrup	Freezer	6 months
Concentrated syrup	Freezer	6–8 months

Freezing stops microbial growth but does not completely kill microorganisms. For this reason, once thawed, the product should be consumed within a short time.

Recommended Formula for Maximum Stability

To achieve the longest shelf life, the following technical parameters are recommended:

Parameter	Target Value
Brix	65
pH	3.2
Filling temperature	75–80 °C
Storage	4–8 °C

When these parameters are achieved, craft fruit syrups can remain stable in the refrigerator for approximately **4–6 months**.

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44. Signs of Syrup Spoilage

If stability cannot be achieved in syrup production, the product may show various microbial or chemical spoilage symptoms. Early detection of these signs is important both for quality control and consumer safety.

Spoilage generally results from the following causes:

- Low sugar ratio
- High pH value
- Insufficient pasteurization
- Poor hygiene
- Improper storage conditions

The table below shows the most common spoilage signs seen in syrup production.

Syrup Spoilage Signs Table

Symptom	Possible Cause	Explanation
Foam formation	Yeast fermentation	Yeast ferments sugars and produces gas
Pressure in bottle	Fermentation	Gas release may be observed when opening the bottle
Cloudiness	Microbial growth	A normally clear syrup becomes cloudy
White layer on surface	Mold growth	Mold may develop on the surface of the syrup
Green or black spots	Mold	Indicates advanced mold growth
Excessively sour odor	Acid bacteria	The flavor profile is spoiled
Alcohol smell	Fermentation	Indicates yeast activity
Darkening of color	Oxidation	Fruit aroma may be degraded
Crystallization	Sugar saturation	Especially seen in high-Brix syrups

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Basic Rules to Prevent Spoilage

To reduce the risk of spoilage in syrup production, the following technical principles should be applied:

1. The Brix value should be kept within **60–65**.

2. The pH value should be between **3.2–3.5**.
3. The product should be pasteurized.
4. Hot filling should be applied.
5. Filtration should be carried out.
6. Bottles should be sterile.
7. The product should be stored in a cool environment.

Applying these rules significantly improves syrup stability.

Detection of Spoiled Product

If a syrup shows the following signs, it should not be consumed:

- Foam formation
- Gas release
- Mold layer
- Bad odor
- Excessive cloudiness

These signs indicate that the product has undergone microbial spoilage.

General Evaluation

When proper production techniques are used, craft fruit syrups can become products that are both aromatically rich and stable. When sugar ratio, pH control, water activity, and hygiene factors are evaluated together, shelf life can be extended significantly.

With appropriate formulation, controlled heat treatment, and proper storage conditions, craft fruit syrups can be produced to remain stable in the refrigerator for several months.

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